GCSE PHYSICS

Higher Tier – Paper 2 style

Topic 4.5: Forces

Practice questions in the style of AQA GCSE Physics (8463). Every question is original; the layout, command words, mark allocations and mark scheme follow the real exam format.

Time allowed	About 240 minutes (roughly 1 minute per mark)
Materials	For this paper you must have: a ruler, a pencil and a calculator.
Instructions	• Use black ink or a black ball-point pen (pencil for diagrams and graphs). • Answer **all** questions. • Answer the questions in the spaces provided. • Do all rough work on a separate sheet. • Cross through any work you do not want to be marked.
Information	• The maximum mark for this paper is **227**. • The marks for questions are shown in brackets. • You are allowed to use a calculator. • You are reminded of the need for good English and clear presentation in your answers. • Questions that ask you to *describe* or *explain* are marked for the science you include, so give each point its own sentence.

Name: _____ Date: _____ Mark: _____ / 227

0 1 This question is about scalar and vector quantities and different types of force.

0 1 . 1 What is the difference between a scalar quantity and a vector quantity? **[2 marks]**

0 1 . 2 Complete the table by writing scalar or vector for each quantity. **[4 marks]**

Quantity	**Scalar or vector?**
Speed	
Velocity	
Mass	
Force	

0 1 . 3 Name **two** contact forces and **two** non-contact forces. **[4 marks]**

0 1 . 4 An astronaut has a mass of 75 kg. Calculate the weight of the astronaut on Earth. Gravitational field strength = 9.8 N/kg. **[3 marks]**

Answer = _____ N

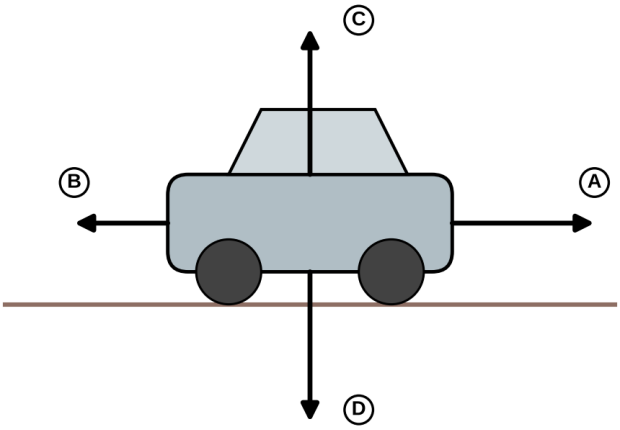
0 1 . 5 The gravitational field strength on the Moon is 1.6 N/kg. Calculate the weight of the astronaut on the Moon and state the astronaut's mass on the Moon. **[3 marks]**

Answer = _____ N

0 1 . 6 State the name of the point where the weight of an object appears to act. **[1 mark]**

_____

0 2 The diagram shows the forces acting on a car of mass 1200 kg travelling along a straight, level road. The driving force is 2600 N and the resistive force is 800 N.



0 2 . 1 Name the forces **A**, **B**, **C** and **D**. **[4 marks]**

_____

_____

_____

_____

0 2 . 2 Calculate the resultant force on the car. **[2 marks]**

Answer = _____ N

0 2 . 3 Write down the equation that links force, mass and acceleration. **[1 mark]**

_____

0 2 . 4 Calculate the acceleration of the car. **[3 marks]**

Answer = _____ m/s^2

0 2 . 5 The driver then keeps the speed steady. State the size of the driving force. Explain your answer. **[2 marks]**

_____

_____

_____

0 2 . 6 State what is meant by the **resultant force**. **[1 mark]**

_____

_____

0 3 This question is about work done and combining forces.

0 3 . 1 Write down the equation that links work done, force and distance. **[1 mark]**

0 3 . 2 A man pushes a trolley with a constant force of 400 N for 12 m. Calculate the work done. **[3 marks]**

Answer = _____ J

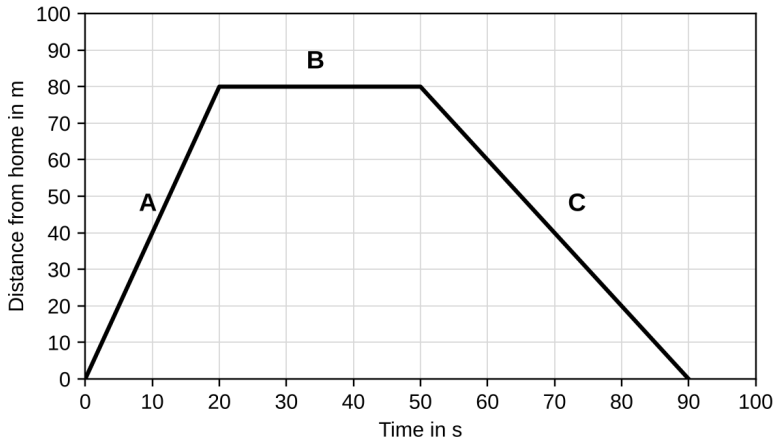
0 3 . 3 State what happens to the energy transferred when a rough surface does work against friction. **[2 marks]**

0 3 . 4 Two forces of 30 N and 40 N act on a box at right angles to each other. Calculate the size of the resultant force. **[3 marks]**

Answer = _____ N

0 3 . 5 A box is pushed with a force of 40 N to the right. A frictional force of 25 N acts to the left. State the resultant force and its direction. **[2 marks]**

0 4 The graph shows the journey of a student from home to a shop and back again.



0 4 . 1 Describe the motion of the student in section **B**. **[1 mark]**

0 4 . 2 Calculate the speed of the student during section **A**. **[3 marks]**

Answer = _____ m/s

0 4 . 3 Calculate the speed in section **C** and explain how it compares with section **A**. **[3 marks]**

Answer = _____ m/s

0 4 . 4 Calculate the average speed for the whole journey. **[3 marks]**

Answer = _____ m/s

0 4 . 5 State the displacement of the student at the end of the journey. **[1 mark]**

_____

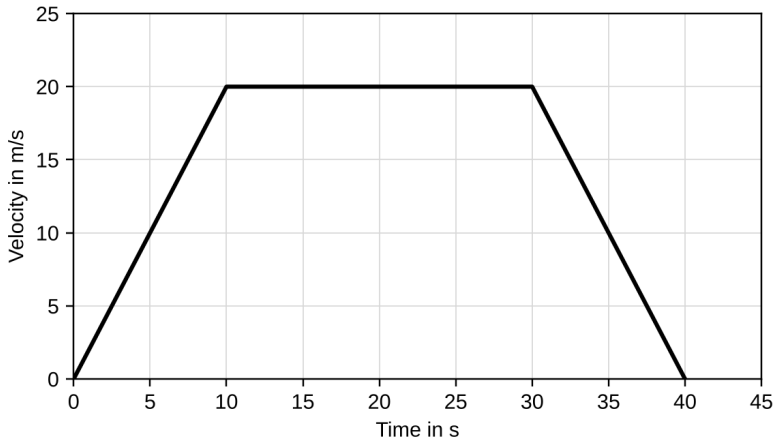
0 4 . 6 Explain the difference between speed and velocity. **[2 marks]**

_____

_____

_____

0 5 The velocity–time graph shows the motion of a car of mass 1000 kg.



0 5 . 1 Calculate the acceleration of the car between 0 and 10 s. **[3 marks]**

Answer = _____ m/s^2

0 5 . 2 Calculate the resultant force on the car between 0 and 10 s. **[2 marks]**

Answer = _____ N

0 5 . 3 Calculate the total distance travelled in the 40 s. **[4 marks]**

Answer = _____ m

0 5 . 4 Calculate the average speed of the car during the 40 s. **[2 marks]**

Answer = _____ m/s

0 5 . 5 Describe the motion of the car between 30 s and 40 s. **[2 marks]**

0 6 This question is about uniform acceleration.

0 6 . 1 Write down the equation that links final velocity, initial velocity, acceleration and distance. **[1 mark]**

0 6 . 2 A cyclist accelerates uniformly from 8.0 m/s to 20 m/s over a distance of 56 m. Calculate the acceleration. **[4 marks]**

Answer = _____ m/s^2

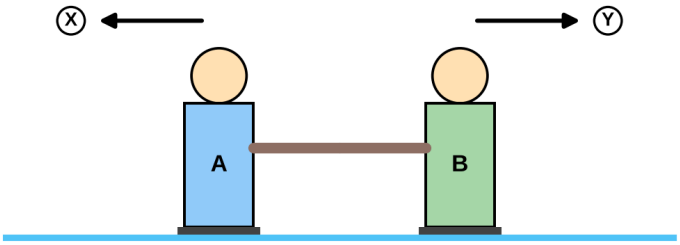
0 6 . 3 A stone is dropped from rest and falls 20 m. Calculate its speed just before it hits the ground, ignoring air resistance. Acceleration due to gravity = 9.8 m/s^2. **[4 marks]**

Answer = _____ m/s

0 6 . 4 Complete the table by giving a typical value for the speed of each. **[3 marks]**

Movement	**Typical speed in m/s**
Person walking	
Person cycling	
Sound in air	

0 7 This question is about Newton's laws of motion.



(skater A is heavier than skater B)

0 7 . 1 State Newton's first law. **[2 marks]**

0 7 . 2 Explain why passengers in a car lurch forwards when the car brakes suddenly. **[3 marks]**

0 7 . 3 What is meant by **inertia**? **[1 mark]**

0 7 . 4 State Newton's third law. **[1 mark]**

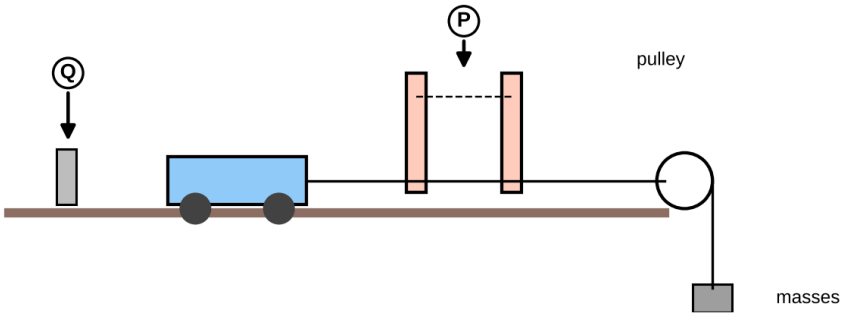
0 7 . 5 Two skaters stand facing each other on ice. Skater **A** (mass 60 kg) pushes skater **B** (mass 40 kg) with a force of 120 N. Which arrow, **X** or **Y**, shows the force on B? **[1 mark]**

0 7 . 6 Calculate the acceleration of each skater. **[4 marks]**

Answer = _____

0 7 . 7 Explain why a book on a table is in equilibrium. Refer to the forces acting on it. **[3 marks]**

0 8 A student investigates how the acceleration of a trolley depends on the resultant force. The diagram shows the apparatus. The student changes the force by moving 10 g masses from the trolley to the hanger. The graph shows her results.



0 8 . 1 Name the part of the apparatus labelled **P** and explain how it is used. **[2 marks]**

0 8 . 2 Explain why the student moves masses from the trolley to the hanger instead of adding extra masses to the hanger. **[2 marks]**

0 8 . 3 State the independent variable and the dependent variable. **[2 marks]**

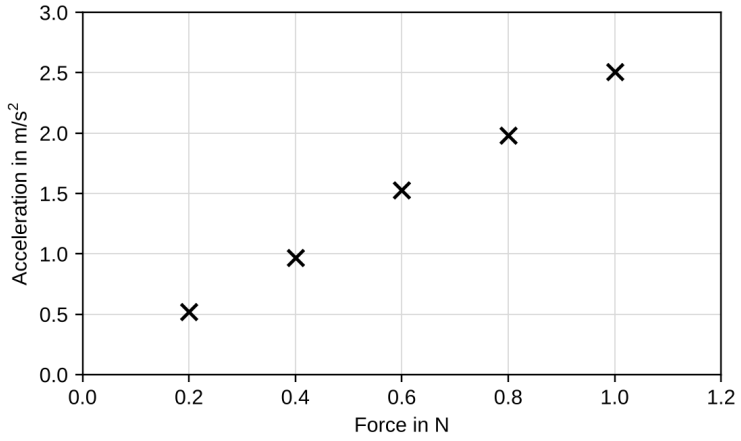
0 8 . 4 Draw a line of best fit on the graph. **[1 mark]**

0 8 . 5 Use the graph to calculate the gradient and the total mass of the trolley and masses. **[4 marks]**

Answer = _____ kg

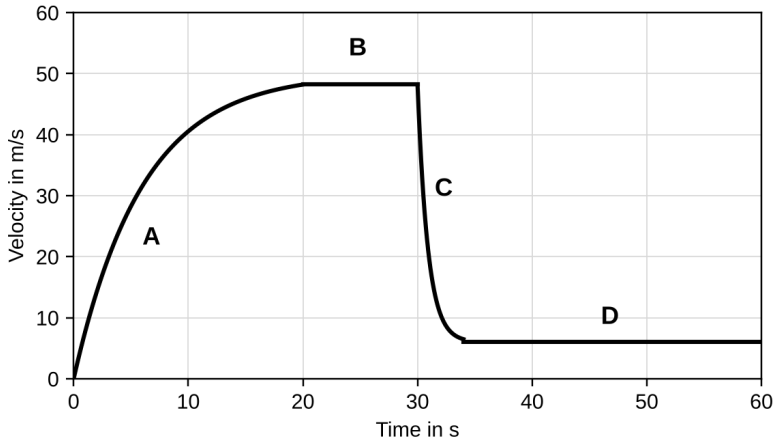
0 8 . 6 State the relationship between force and acceleration shown by the graph. **[1 mark]**

0 9 The student's results are plotted below.



0 9 . 1 Why is it not possible to be sure the line passes through the origin? Suggest a reason for any intercept on the acceleration axis. **[2 marks]**

1 0 A skydiver jumps from a plane and later opens a parachute. The graph shows how the velocity of the skydiver changes with time.



1 0 . 1 Describe and explain the motion of the skydiver in section **A**. **[3 marks]**

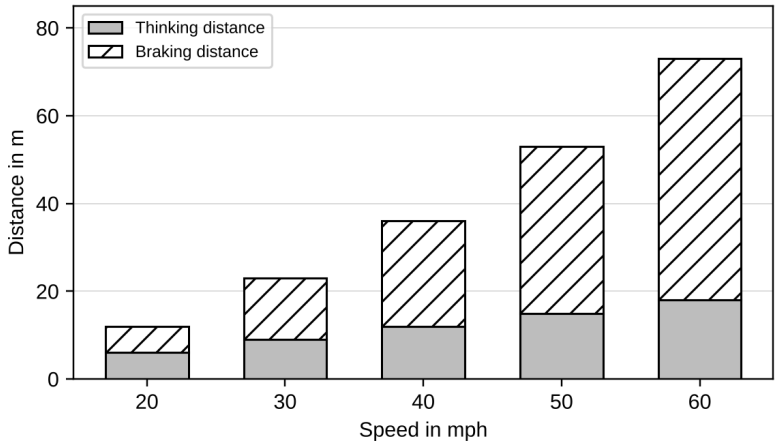
1 0 . 2 In section **B** the skydiver is falling at a constant velocity. State the name for this velocity and the size of the air resistance, given that the skydiver's weight is 800 N. **[3 marks]**

1 0 . 3 The skydiver opens the parachute at 30 s. Explain the motion in section **C**. **[3 marks]**

1 0 . 4 Explain why the skydiver reaches a lower terminal velocity in section **D**. **[2 marks]**

1 0 . 5 Use the graph to find the terminal velocity in section **D**. **[1 mark]**

1 1 The bar chart shows the thinking distance and braking distance of a car at different speeds.



1 1 . 1 What is meant by the **stopping distance**? **[1 mark]**

1 1 . 2 Use the chart to find the stopping distance at 40 mph. **[1 mark]**

1 1 . 3 Describe how the thinking distance changes with speed. **[2 marks]**

1 1 . 4 Calculate the percentage increase in stopping distance when the speed increases from 30 mph to 60 mph. **[4 marks]**

Answer = _____ %

1 1 . 5 Give **two** factors that increase thinking distance. **[2 marks]**

1 1 . 6 Give **two** factors that increase braking distance. **[2 marks]**

1 1 . 7 A driver has a reaction time of 0.70 s. The car is travelling at 20 m/s. Calculate the thinking distance. **[3 marks]**

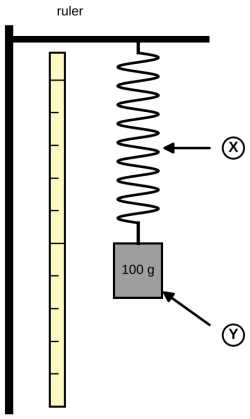
Answer = _____ m

1 1 . 8 A car of mass 800 kg travels at 20 m/s. Calculate its kinetic energy and, given a braking distance of 40 m, calculate the average braking force. **[5 marks]**

Answer = _____ N

1 1 . 9 Describe how a ruler-drop test could be used to measure reaction time. **[3 marks]**

1 2 A student investigates how the extension of a spring depends on the force applied. The diagram shows the apparatus.



1 2 . 1 Name the parts labelled **X** and **Y**. **[2 marks]**

1 2 . 2 Describe how the student should measure the extension of the spring. **[3 marks]**

1 2 . 3 State **one** way to reduce reading errors when using the ruler. **[1 mark]**

1 2 . 4 The student plots force against extension, as shown. Calculate the spring constant of the spring in N/m. **[4 marks]**

Answer = _____ N/m

1 2 . 5 What is the name for point **P**? **[1 mark]**

1 2 . 6 Calculate the extension when a force of 3.0 N is applied. **[3 marks]**

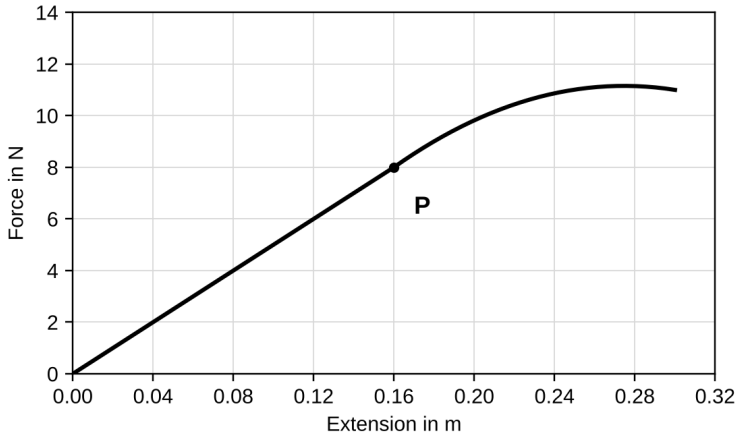
Answer = _____ m

1 2 . 7 Calculate the energy stored in the spring when it is stretched by 0.16 m. **[3 marks]**

Answer = _____ J

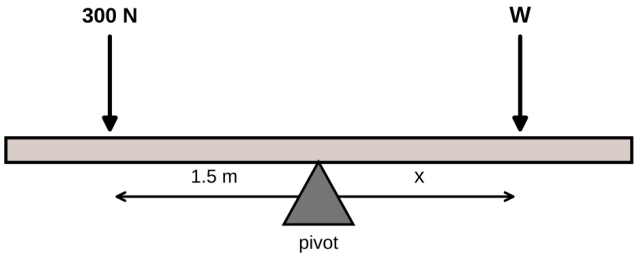
1 2 . 8 Explain what happens to the spring if the force is increased beyond point **P** and then removed. **[2 marks]**

1 3 The force–extension graph for the spring is shown.



1 3 . 1 State the relationship between force and extension for the first part of the graph. **[1 mark]**

1 4 A child of weight 300 N sits 1.5 m from the pivot of a seesaw. An adult of weight **W** sits on the other side at a distance **x** from the pivot. The seesaw is balanced.



1 4 . 1 Write down the equation that links moment, force and distance. **[1 mark]**

1 4 . 2 Calculate the moment of the child about the pivot. **[3 marks]**

Answer = _____ N m

1 4 . 3 The adult weighs 600 N. Calculate the distance **x**. **[3 marks]**

Answer = _____ m

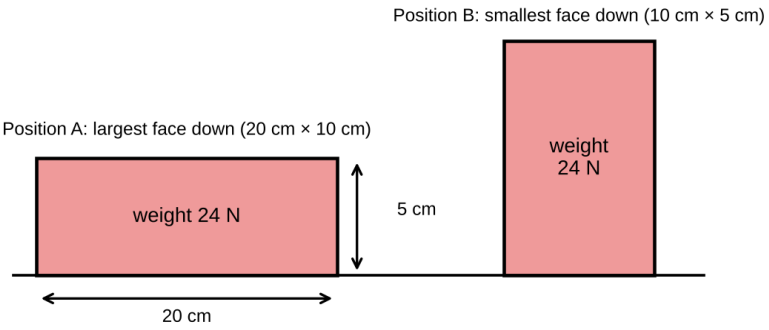
1 4 . 4 A man uses a spanner of length 0.35 m to apply a force of 40 N. Calculate the moment. **[3 marks]**

Answer = _____ N m

1 4 . 5 Explain why using a longer spanner makes it easier to undo a tight nut. **[2 marks]**

1 4 . 6 A gear **A** with 20 teeth turns at 60 rotations per minute and drives gear **B** with 40 teeth. Calculate the rate at which B turns and its direction relative to A. **[3 marks]**

1 5 This question is about pressure.



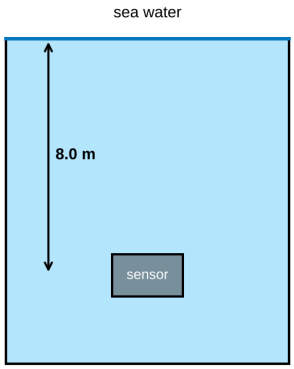
1 5 . 1 Write down the equation that links pressure, force and area. **[1 mark]**

1 5 . 2 A brick weighs 24 N. Calculate the pressure on the ground in position **A** (area 0.020 m^2) and position **B** (area 0.0050 m^2). **[4 marks]**

Answer = _____ Pa

1 5 . 3 Explain why a drawing pin has a sharp point and a wide head. **[2 marks]**

1 6 A sensor is lowered into the sea to a depth of 8.0 m. The density of sea water is 1030 kg/m^3. Gravitational field strength = 9.8 N/kg.



1 6 . 1 Write down the equation that links pressure due to a column of liquid, height, density and gravitational field strength. **[1 mark]**

1 6 . 2 Calculate the pressure due to the water on the sensor. **[3 marks]**

Answer = _____ Pa

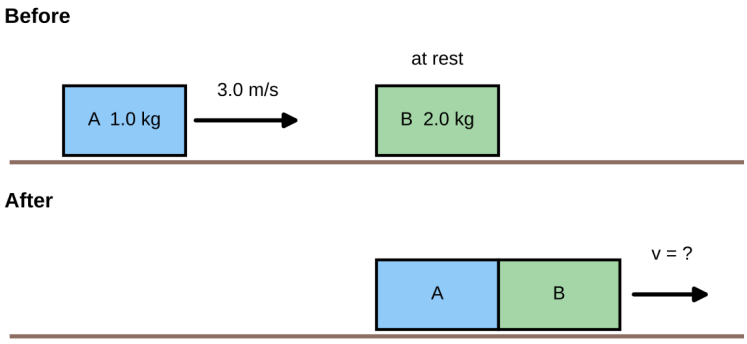
1 6 . 3 Explain why the pressure in a liquid increases with depth. **[2 marks]**

1 6 . 4 Explain what causes upthrust on an object in water. **[3 marks]**

16.5 Explain why a boat made of steel floats although a steel block sinks. **[2 marks]**

16.6 Explain why atmospheric pressure decreases with increasing height above the Earth's surface. **[2 marks]**

17 This question is about momentum.



17.1 Write down the equation that links momentum, mass and velocity. **[1 mark]**

17.2 Calculate the momentum of a 1000 kg car travelling at 20 m/s. **[3 marks]**

Answer = _____ kg m/s

1 7 . 3 Trolley **A** (1.0 kg) moves at 3.0 m/s and collides with a stationary trolley **B** (2.0 kg). They stick together. Calculate the velocity of the trolleys after the collision. **[4 marks]**

Answer = _____ m/s

1 7 . 4 State the principle of conservation of momentum. **[2 marks]**

1 7 . 5 A car of mass 1000 kg travelling at 20 m/s stops in 0.50 s in a collision. Calculate the average force on the car. **[4 marks]**

Answer = _____ N

1 7 . 6 Explain how a crumple zone reduces the injuries to the people in a car. **[3 marks]**

1 8 Read the question and answer in full sentences.

1 8 . 1 *Describe an investigation to find out how the extension of a spring depends on the force applied. Include the apparatus used, the measurements taken and how the results are analysed. **[6 marks]**

1 8 . 2 *A driver says: 'If I drive slowly, my stopping distance will always be short.' Evaluate this claim by explaining the factors that affect stopping distance. **[6 marks]**

END OF QUESTIONS

MARK SCHEME – Topic 4.5: Forces

How to use this mark scheme: mark your own work in a different colour. A semicolon (;) separates marking points and each is worth one mark unless stated. A slash (/) separates alternative answers that are equally acceptable. Words in brackets are not needed to gain the mark. **Allow** = acceptable; **Ignore** = neither credited nor penalised; **Do not accept** = contradicts or spoils the answer. Marks are for the *science*, not the wording.

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.1	scalar has magnitude (size) only ; vector has magnitude and direction ;		2	AO1 4.5.1.1
01.2	speed – scalar ; velocity – vector ; mass – scalar ; force – vector ;		4	AO1 4.5.1.1
01.3	contact: friction / tension / air resistance / normal contact force (any two) ;; non-contact: gravitational / electrostatic / magnetic (any two) ;;	One mark each, max 2 per type.	4	AO1 4.5.1.1
01.4	W = m g ; = 75 × 9.8 ; 735 (N) ;		3	AO2 4.5.1.3
01.5	W = 75 × 1.6 = 120 (N) ; mass = 75 kg ; mass does not change (only weight changes) ;		3	AO2 4.5.1.3
01.6	centre of mass		1	AO1 4.5.1.3
02.1	A: driving force / thrust ; B: air resistance / drag / friction ; C: normal contact force (reaction) ; D: weight ;		4	AO1 4.5.1.1
02.2	2600 − 800 ; 1800 (N) ;		2	AO2 4.5.1.4
02.3	force = mass × acceleration		1	AO1 4.5.4.1
02.4	a = F ÷ m ; = 1800 ÷ 1200 ; 1.5 (m/s^2) ;		3	AO2 4.5.4.1
02.5	800 (N) ; resultant force is zero so the speed is constant (forces balanced) ;		2	AO2 4.5.4.1
02.6	single force that has the same effect as all the forces acting together		1	AO1 4.5.1.4
03.1	work done = force × distance (moved along the line of action of the force)		1	AO1 4.5.2
03.2	W = F s ; = 400 × 12 ; 4800 (J) ;		3	AO2 4.5.2
03.3	energy transferred to thermal store ; surface and objects become warmer ;		2	AO1 4.5.2

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.4	$R^2 = 30^2 + 40^2$; = 2500 ; 50 (N) ;	Allow scale drawing: 50 N ± 2 N.	3	AO2 4.5.1.4
03.5	15 N ; to the right ;		2	AO2 4.5.1.4
04.1	stationary / at rest (at the shop)		1	AO3 4.5.6.1
04.2	speed = distance ÷ time ; 80 ÷ 20 ; 4.0 (m/s) ;		3	AO3 4.5.6.1
04.3	80 ÷ 40 = 2.0 m/s ; slower (half the speed) ; line is less steep ;		3	AO3 4.5.6.1
04.4	total distance = 160 m ; 160 ÷ 90 ; 1.8 (m/s) ;	Allow 1.78.	3	AO3 4.5.6.1
04.5	0 (m)		1	AO2 4.5.6.1
04.6	speed is a scalar (no direction) ; velocity is a vector (speed in a given direction) ;		2	AO1 4.5.6.1
05.1	a = Δv ÷ t ; 20 ÷ 10 ; 2.0 (m/s^2) ;		3	AO3 4.5.6.2.2
05.2	F = m a = 1000 × 2.0 ; 2000 (N) ;		2	AO2 4.5.4.1
05.3	distance = area under graph ; 0–10 s: ½ × 10 × 20 = 100 m ; 10–30 s: 20 × 20 = 400 m ; 30–40 s: 100 m, total 600 (m) ;		4	AO3 4.5.6.2.2
05.4	600 ÷ 40 ; 15 (m/s) ;	Allow ecf.	2	AO2 4.5.6.1
05.5	decelerates / slows down ; uniformly, to rest (−2 m/s^2) ;		2	AO3 4.5.6.2.2
06.1	$(\text{final velocity})^2 - (\text{initial velocity})^2 = 2 \times$ acceleration × distance		1	AO1 4.5.6.2.2
06.2	$20^2 - 8^2 = 2 \times a \times 56$; 336 = 112 a ; a = 336 ÷ 112 ; 3.0 (m/s^2) ;		4	AO2 4.5.6.2.2
06.3	$v^2 - 0 = 2 \times 9.8 \times 20$; $v^2 = 392$; v = √392 ; 19.8 (m/s) ;	Allow 20 m/s.	4	AO2 4.5.6.2.2
06.4	walking 1.5 m/s ; cycling 6 m/s ; sound in air 330 m/s ;	Allow 1–2, 4–8, 300–340.	3	AO1 4.5.6.1

Question	Answers	Extra information	Mark	AO / Spec. Ref.
07.1	an object stays at rest or moves at constant velocity ; unless acted on by a resultant force ;		2	AO1 4.5.4.1
07.2	passengers' bodies continue to move at the same speed (inertia) ; no resultant force on them (until the seat belt acts) ; while the car slows down ;		3	AO2 4.5.4.3
07.3	the tendency of an object to stay at rest or keep moving at constant velocity		1	AO1 4.5.4.3
07.4	when two objects interact, the forces they exert on each other are equal and opposite		1	AO1 4.5.4.2
07.5	Y		1	AO2 4.5.4.2
07.6	A: a = 120 ÷ 60 ; 2.0 (m/s^2) ; B: a = 120 ÷ 40 ; 3.0 (m/s^2) ;	The force on A is also 120 N (Newton's third law).	4	AO2 4.5.4.2
07.7	weight acts downwards ; normal contact force from the table acts upwards ; forces are equal and opposite so resultant force is zero ;		3	AO2 4.5.4.1
08.1	light gate / (light) sensor ; measures the speed / times the card passing to find acceleration ;		2	AO1 4.5.4.1 RP
08.2	keeps the total mass (of the system) constant ; so only the force changes ;		2	AO3 4.5.4.1 RP
08.3	independent: force / weight on the hanger ; dependent: acceleration ;		2	AO3 4.5.4.1 RP
08.4	straight line through (or near) the origin and the points ;		1	AO3 WS 3.3
08.5	gradient = Δa ÷ ΔF = 2.5 ÷ 1.0 ; = 2.5 (m/s^2 per N) ; mass = 1 ÷ gradient ; 0.40 (kg) ;	Allow 0.38–0.42.	4	AO3 4.5.4.1 RP
08.6	acceleration is directly proportional to force		1	AO3 4.5.4.1
09.1	friction acts on the trolley ; so a small force is needed before it accelerates / the track should be tilted to compensate ;		2	AO3 4.5.4.1 RP
10.1	accelerates / velocity increases ; weight is greater than air resistance ; resultant force downwards (acceleration decreases as air resistance increases) ;		3	AO3 4.5.4.3
10.2	terminal velocity ; 800 (N) ; weight = air resistance (forces balanced) ;		3	AO2 4.5.4.3

Question	Answers	Extra information	Mark	AO / Spec. Ref.
10.3	air resistance increases suddenly (parachute has a greater surface area) ; air resistance > weight so a resultant force upwards ; the skydiver decelerates ;		3	AO3 4.5.4.3
10.4	parachute gives greater surface area / larger drag for a given speed ; so forces balance at a lower speed ;		2	AO3 4.5.4.3
10.5	6 (m/s)		1	AO3 WS 3.1
11.1	thinking distance + braking distance		1	AO1 4.5.6.3
11.2	36 (m)		1	AO3 WS 3.1
11.3	increases ; directly proportional / doubles when speed doubles ;		2	AO3 4.5.6.3
11.4	stopping at 30 = 23 m ; stopping at 60 = 73 m ; (73 − 23) ÷ 23 ; 217 (%) ;	Allow 217–220.	4	AO3 4.5.6.3
11.5	tiredness ; alcohol / drugs ; distractions (e.g. mobile phone) ; higher speed ;	Any two.	2	AO1 4.5.6.3
11.6	wet / icy road ; worn tyres / worn brakes ; greater mass of car ; higher speed ;	Any two.	2	AO1 4.5.6.3
11.7	d = v t ; = 20 × 0.70 ; 14 (m) ;		3	AO2 4.5.6.3
11.8	$E_k = 0.5 \times 800 \times 20^2$; 160 000 (J) ; work done by brakes = kinetic energy ; F = E ÷ d = 160 000 ÷ 40 ; 4000 (N) ;		5	AO3 4.5.6.3
11.9	one person holds a ruler vertically, the other has fingers ready at the zero mark ; ruler is dropped without warning ; distance fallen before it is caught is measured and converted to time (repeat and average) ;		3	AO1 4.5.6.3
12.1	X: spring ; Y: (slotted) mass / weight ;		2	AO1 4.5.3 RP
12.2	measure the natural (unstretched) length with the ruler ; measure the length with each mass added ; extension = new length − natural length ;		3	AO1 4.5.3 RP

Question	Answers	Extra information	Mark	AO / Spec. Ref.
12.3	read at eye level / use a set square / fix pointer to spring ;		1	AO3 4.5.3 RP
12.4	gradient of straight section ; 8 ÷ 0.16 ; 50 ; N/m ;		4	AO3 4.5.3
12.5	limit of proportionality		1	AO1 4.5.3
12.6	F = k e ; e = 3.0 ÷ 50 ; 0.060 (m) ;		3	AO2 4.5.3
12.7	$E = 0.5 \text{ k } e^2$; $= 0.5 \times 50 \times 0.16^2$; 0.64 (J) ;		3	AO2 4.5.3
12.8	extension is not proportional to force beyond P ; the spring does not return to its original length (inelastic deformation) ;		2	AO2 4.5.3
13.1	directly proportional / linear (F = k e)		1	AO3 4.5.3
14.1	moment = force × perpendicular distance from the pivot		1	AO1 4.5.3
14.2	M = F d ; = 300 × 1.5 ; 450 (N m) ;		3	AO2 4.5.3
14.3	clockwise moment = anticlockwise moment ; 600 × x = 450 ; 0.75 (m) ;		3	AO2 4.5.3
14.4	M = 40 × 0.35 ; 14 (N m) ; unit ;		3	AO2 4.5.3
14.5	moment = force × distance ; larger distance so a smaller force produces the same moment ;		2	AO2 4.5.3
14.6	B has twice the teeth so turns at half speed ; 30 rotations per minute ; in the opposite direction ;		3	AO3 4.5.3
15.1	pressure = force ÷ area (normal to the surface)		1	AO1 4.5.5.1
15.2	A: 24 ÷ 0.020 ; 1200 (Pa) ; B: 24 ÷ 0.0050 ; 4800 (Pa) ;		4	AO2 4.5.5.1
15.3	small area at the point gives a large pressure (to push into the wall) ; large area at the head gives a smaller pressure on the thumb ;		2	AO2 4.5.5.1
16.1	pressure = height × density × gravitational field strength		1	AO1 4.5.5.2

Question	Answers	Extra information	Mark	AO / Spec. Ref.
16.2	p = h ρ g ; = 8.0 × 1030 × 9.8 ; 80 752 (Pa) / 81 000 ;		3	AO2 4.5.5.2
16.3	greater weight of liquid above ; more particle collisions / more water above pressing down ;		2	AO2 4.5.5.2
16.4	pressure increases with depth ; so the pressure on the bottom surface is greater than on the top ; giving a resultant (upward) force ;		3	AO2 4.5.5.2
16.5	the boat displaces a large volume of water ; so upthrust equals its weight / average density less than water ;		2	AO3 4.5.5.2
16.6	fewer air molecules / thinner air above ; so less weight of air above (fewer collisions) ;		2	AO2 4.5.5.3
17.1	momentum = mass × velocity		1	AO1 4.5.7.1
17.2	p = m v ; = 1000 × 20 ; 20 000 (kg m/s) ;		3	AO2 4.5.7.1
17.3	momentum before = 1.0 × 3.0 = 3.0 kg m/s ; momentum after = (1.0 + 2.0) × v ; 3.0 = 3.0 v ; 1.0 (m/s) ;		4	AO2 4.5.7.2
17.4	in a closed system the total momentum before an event ; equals the total momentum after ;		2	AO1 4.5.7.2
17.5	change in momentum = 20 000 ; F = Δp ÷ t ; 20 000 ÷ 0.50 ; 40 000 (N) ;		4	AO3 4.5.7.3
17.6	increases the time taken for the car to stop ; so the rate of change of momentum is smaller ; so the force on the people is smaller ;		3	AO2 4.5.7.3

Question	Answers	Extra information	Mark	AO / Spec. Ref.
18.1	**Level 3 (5–6 marks)**: A clear, detailed and logically ordered answer that covers all of the key areas, with correct scientific reasoning linking the ideas together. **Level 2 (3–4 marks)**: Some of the key areas are covered with some correct reasoning, but the answer lacks detail, balance or logical order in places. **Level 1 (1–2 marks)**: Simple, separate statements with little or no linking or reasoning. **0 marks**: No relevant content. **Indicative content** (credit any other correct science): • Level 3 (5–6): clear, logical method with correct apparatus, measurements, control of variables and a full analysis. • Level 2 (3–4): mostly correct method with some detail missing. • Level 1 (1–2): simple, incomplete statements. • Indicative content: clamp stand, spring, ruler, masses (weights) on a hanger; measure natural length; add masses one at a time; measure new length; calculate extension; weight = mass × g; repeat; plot force against extension; straight line through origin shows proportionality; gradient = spring constant; stop before limit of proportionality.		6	AO3 4.5.3 RP
18.2	**Level 3 (5–6 marks)**: A clear, detailed and logically ordered answer that covers all of the key areas, with correct scientific reasoning linking the ideas together. **Level 2 (3–4 marks)**: Some of the key areas are covered with some correct reasoning, but the answer lacks detail, balance or logical order in places. **Level 1 (1–2 marks)**: Simple, separate statements with little or no linking or reasoning. **0 marks**: No relevant content. **Indicative content** (credit any other correct science): • Level 3 (5–6): balanced evaluation covering thinking and braking distances with several factors and a conclusion. • Level 2 (3–4): some factors explained. • Level 1 (1–2): simple statements. • Indicative content: lower speed gives shorter thinking and braking distances (speed squared in braking); but thinking distance also depends on reaction time (tiredness, alcohol, distractions); braking distance depends on road conditions, tyre and brake condition, mass; ice or wet roads may give a long distance even at low speed; so the claim is only partly true.		6	AO3 4.5.6.3

Where marks are lost in Topic 4.5

Use this page after marking. Tick the ones that cost you marks, then redo those questions closed-book.

- ☐ **Free-body diagrams:** draw arrows from the centre of the object; length shows size. Label every force and say what it is (weight, normal contact force, driving force, drag).
- ☐ **Resultant force** = total of forces along a line (subtract opposite directions). If it is zero the object stays at rest or moves at constant velocity.
- ☐ **F = m a:** mass in kg, force in N. A bigger mass gives a smaller acceleration for the same force.
- ☐ **Graphs:** distance–time gradient = speed; velocity–time gradient = acceleration; area under a velocity–time graph = distance. Show your triangle or rectangle calculations.
- ☐ **Terminal velocity:** weight = drag. Mention the 3 stages: acceleration → drag increases → forces balanced. A parachute increases the surface area.
- ☐ **Stopping distance** = thinking + braking. Thinking distance depends on reaction time (and speed); braking distance depends on speed, mass, road and brakes.
- ☐ **Hooke's law:** $F = k\,e$, $E = \frac{1}{2}\,k\,e^2$. Use extension (not total length) and metres.
- ☐ **Moments:** measure the distance **perpendicular** to the line of the force. Anticlockwise moments = clockwise moments when balanced.
- ☐ **Momentum:** $p = m\,v$. In a collision, total momentum before = after. Longer collision time → smaller force (crumple zones, airbags).
- ☐ **Pressure:** $p = F \div A$. In liquids $p = h\,\rho\,g$ — more depth, more pressure; upthrust is caused by the pressure difference.